

Machine Learning and Organizational Adaptation: How Digital Systems Learn

Executive Summary

Machine learning gives organizations the ability to build systems that **update their own models from data** — automating the learning process itself. Under Active Inference, ML is formalized model updating: algorithms that revise their generative models based on new evidence. This module covers ML deployment in organizations, the model lifecycle, data feedback loops, and the challenge of maintaining ML systems as the world changes.

Learning Objectives

1. Frame **machine learning** as automated model updating within the Active Inference framework
 2. Understand the **ML lifecycle** — from data collection through model deployment and monitoring
 3. Design **feedback loops** that keep ML models aligned with organizational objectives
 4. Address **model drift** — when models degrade because the world changes
 5. Navigate **MLOps** — the operational infrastructure for managing ML at scale
-

Key Concepts

1. ML as Automated Inference

ML Function	Active Inference Translation	Organizational Application
Supervised learning	Learning the mapping from observations to hidden states	Customer churn prediction, fraud detection
Unsupervised learning	Discovering structure in observations without labeled states	Customer segmentation, anomaly detection
Reinforcement learning	Learning optimal policies through action-outcome feedback	Dynamic pricing, resource allocation
Deep learning	Hierarchical generative models with many layers	Image recognition, NLP, recommendation

2. The ML Model Lifecycle

Data Collection → Data Preparation → Model Training → Evaluation → Deployment

Each stage introduces potential failure:

- **Data collection:** Biased or incomplete data
- **Training:** Overfitting, wrong objectives
- **Deployment:** Integration failures, latency issues
- **Monitoring:** Undetected drift, silent failures

3. Model Drift and Retraining

Case Study — Zillow's iBuying Algorithm: Zillow's "Zestimate" algorithm, which predicted home prices, worked well in stable markets but failed catastrophically when the 2021 housing market shifted rapidly. The model was trained on historical patterns that no longer held — a textbook case of model drift. Zillow lost \$381M and shut down the iBuying division. The

lesson: ML models are not "set and forget" — they require continuous monitoring and retraining as the world changes.

4. Organizational ML Maturity

Level	Characteristics	Challenge
Ad hoc	Individual data scientists build models	No standards, no reproducibility
Repeatable	Standard processes for model development	Limited scale, manual deployment
Managed	MLOps infrastructure, automated pipelines	Organizational integration
Optimized	ML embedded in core business processes	Maintaining alignment as scale increases

Cross-References

- For organizational learning, see [Organizational Systems: Learning](#)
 - For collective memory, see [Collective Intelligence: Learning](#)
 - For adaptive strategy, see [Strategic Modeling: Learning](#)
-

Summary

Concept	Digital Transformation Meaning
Machine learning	Automated model updating from data
ML lifecycle	Data → training → deployment → monitoring → retraining
Model drift	Model degradation as the world changes
MLOps	Operational infrastructure for managing ML at scale
Feedback loops	Mechanisms for keeping ML models aligned with objectives

References

- Sculley, D. et al. (2015). Hidden technical debt in machine learning systems. *NeurIPS*.
- Paleyes, A. et al. (2022). Challenges in deploying machine learning. *ACM Computing Surveys*, 55(6).